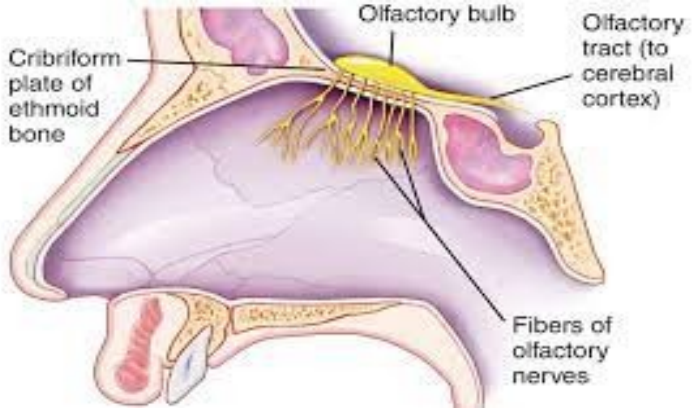
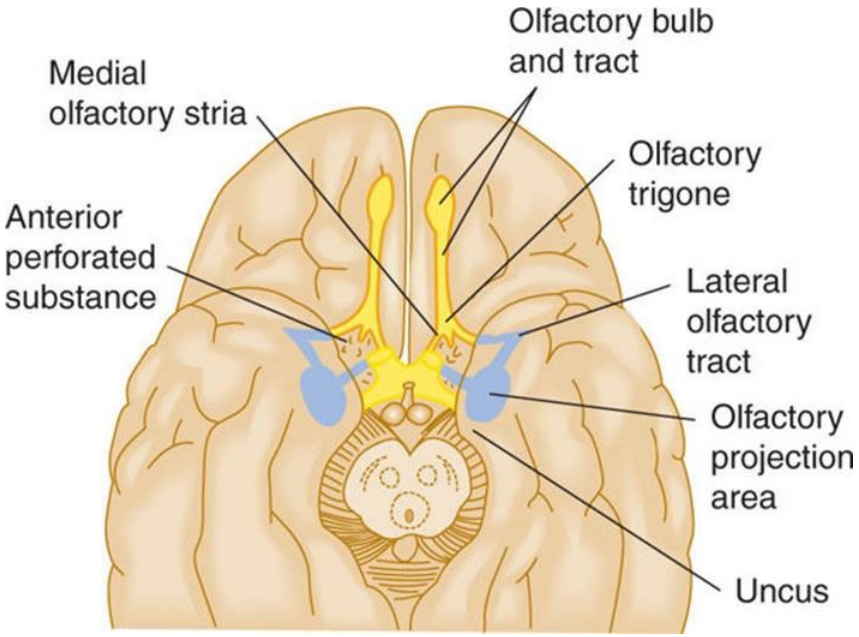
Cranial nerves 1, 3, 4, & 6.

Olfactory nerve (1st):

- It is sensory in nature.
- It arises from neuroepithelium of upper one-third of nasal cavity.
- This neuroepithelium contains bipolar neurons.
- 18-20 olfactory nerve fibers arise from these bipolar neurons and pass through cribriform plate of ethmoid bone.
- These olfactory nerve fibers synapse with the cells of olfactory bulb.
- The olfactory tract travels posteriorly on the inferior surface of frontal lobe and ends in the olfactory trigone.
- The olfactory trigone is a triangular widening of the terminal olfactory tract located superior to the anterior clinoid process and to the anterior perforated substance.





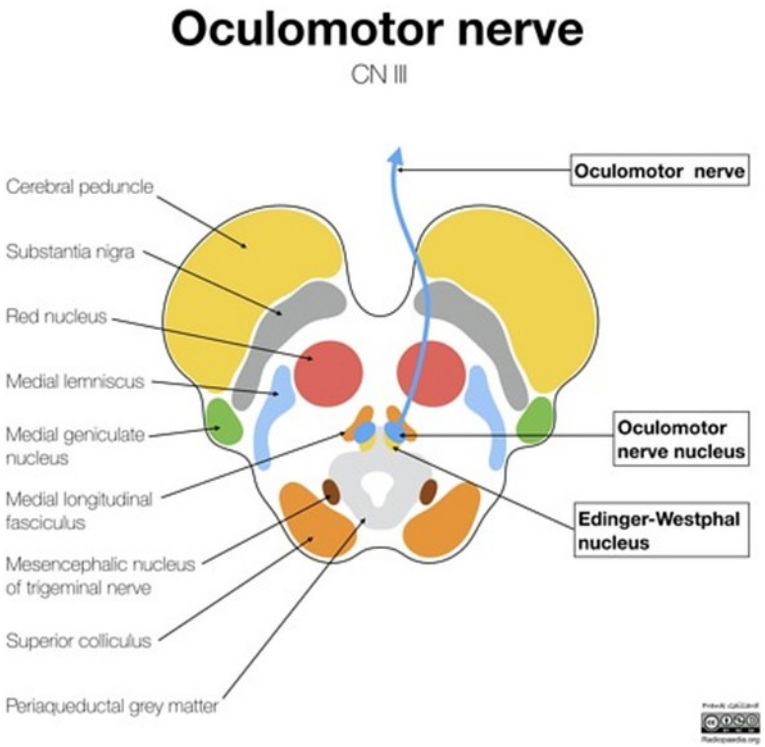
Oculomotor nerve (3rd):

- It is a motor nerve.
- It supplies extraocular as well as intraocular muscles.
- **Nuclei:**

I. Oculomotor nucleus which is located in the central grey matter of midbrain at the level of superior colliculus.

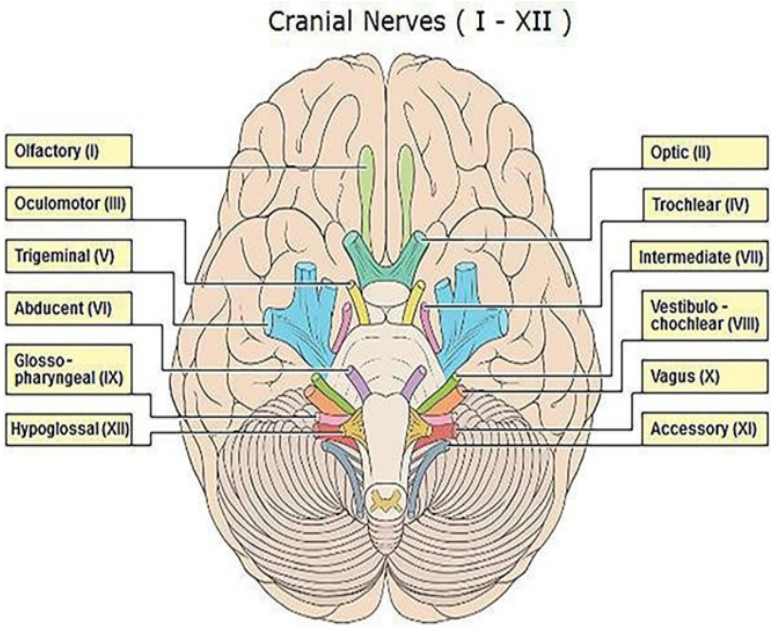
II. Edinger Westphal nucleus is a part of oculomotor nuclear complex from which fibers for constrictor pupillae and ciliary muscles arise.

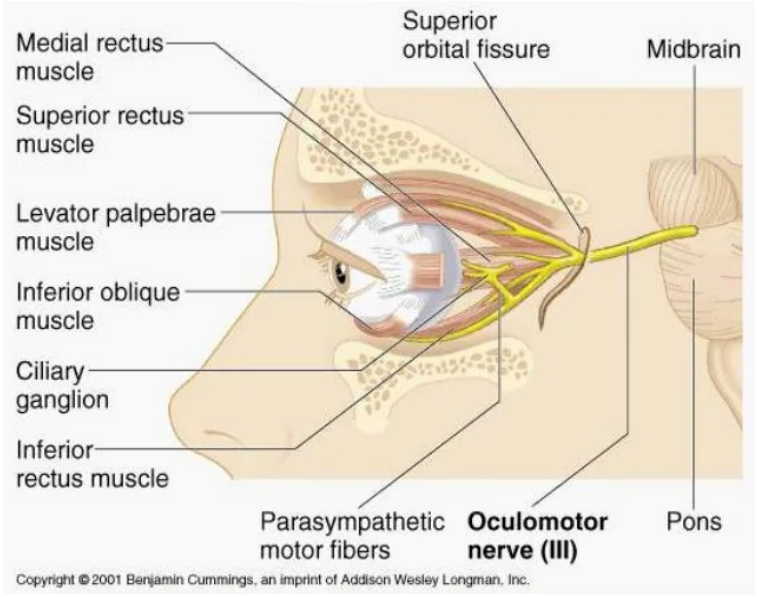
- The oculuomotor nucleus is connected to pretectal nuclei for the light reflex.



Course:

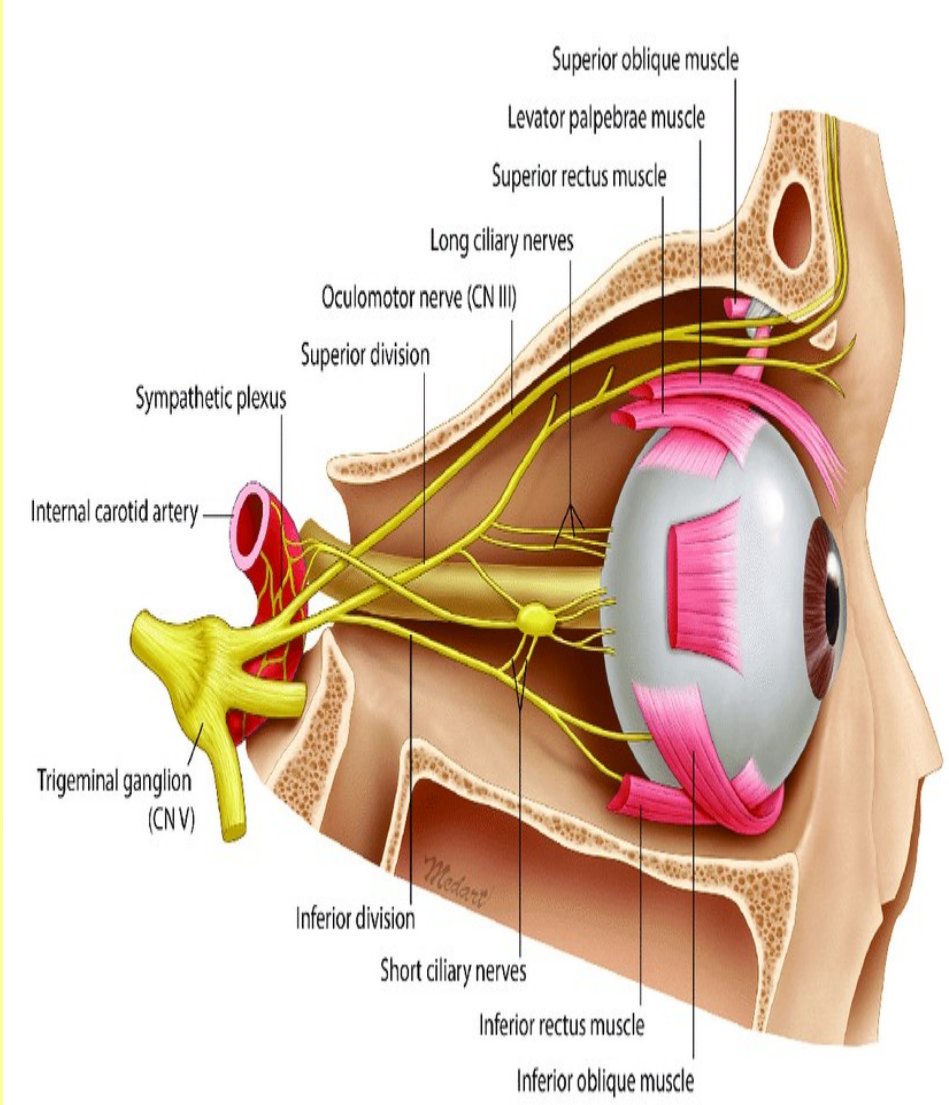
- At the base of brain, the nerve is attached to oculomotor sulcus medial to crus cerebri.
- Then it runs through interpeduncular fossa to reach the cavernous sinus.
- It runs in the lateral wall of cavernous sinus above trochlear nerve.
- It splits into an upper smaller division and a lower larger division in the anterior part of cavernous sinus.
- Then enters the orbit through superior orbital fissure and supplies the extraocular muscles through upper and lower divisions.





CILIARY GANGLION

- **SITE:** in the fat **between** optic nerve & lateral rectus muscle
- **NUCLEUS:** **Edinger Westphal** of occulomotor (midbrain)
- **PREGANGLIONIC FIBERS:** along occulomotor through **nerve to inferior oblique**
- **POSTGANGLIONC FIBERS:** along short ciliary nerves to **sphincter pupillae & ciliary muscles**



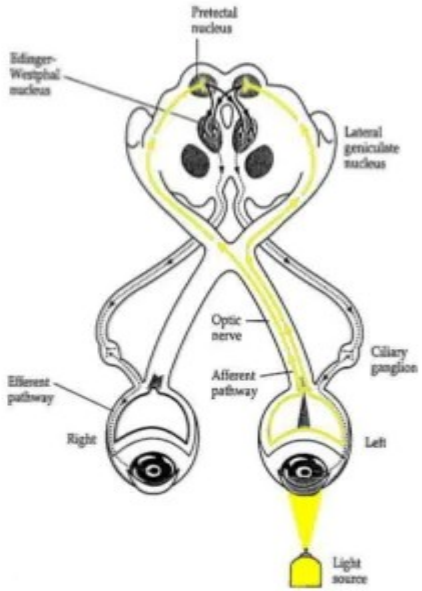
Pupillary Reflexes



Light reflex:

- **Pupillary light reflex (PLR)** : that controls the diameter of the pupil, in response to the intensity of light that falls on the retinal ganglion cells of the retina in the back of the eye.

- **Stimulus:** direct light (torch).
- **Affrent:** optic nerve fibers(**CNII**).
- **Center:** B/S
- **Eefrent:** short ciliary nerve fibers (**CNIII**)
- **Effector organ:** Pupillary constrictor muscle.
- **General pathway (**important**)**
  Ganglion cell ⟶ optic nerve ⟶ pretectal nuclei⟶ Edinger-Westphal nucleus⟶ the ciliary ganglion⟶ short ciliary nerve fiber⟶ Pupillary constrictor.



Applied anatomy:

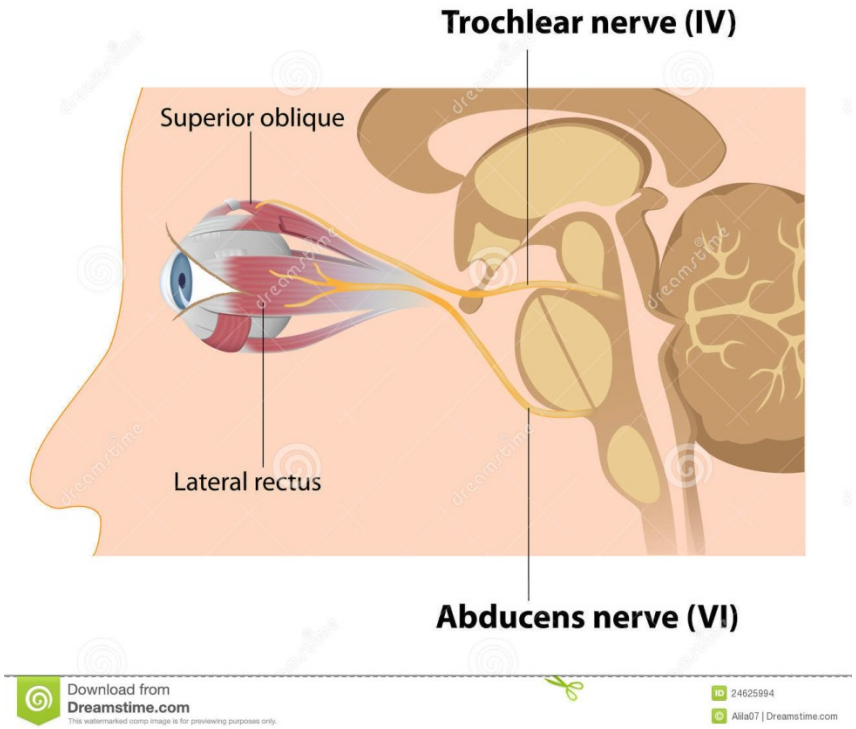
- Complete paralysis of 3rd nerve results in:

I. Ptosis (drooping of upper eyelid)
II. Lateral squint
III. Dilatation of pupil
IV. Loss of accomodation
V. Mild proptosis (forward projection of eyeball)
VI. Diplopia

- **Squint:** it occurs as a result of weakness/paralysis of any extraocular muscle in which both eyeballs are not aligned in the same direction.
- Squint may be congenital or acquired.

Trochlear nerve (4th):

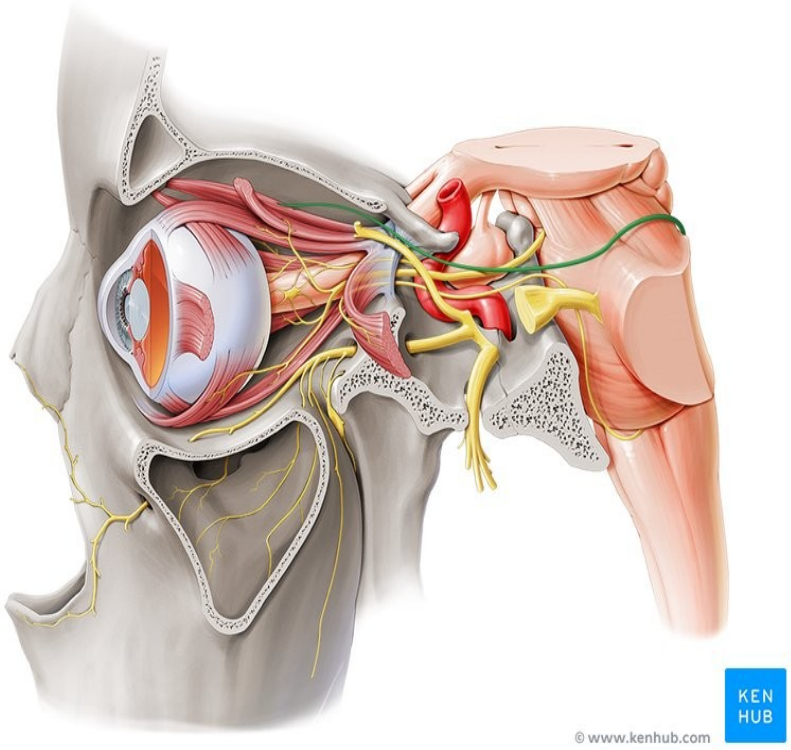
- It is motor and supplies superior oblique muscle of the eyeball.
- **Nucleus:** trochlear nucleus is located in the central grey matter of midbrain at the level of inferior colliculus.



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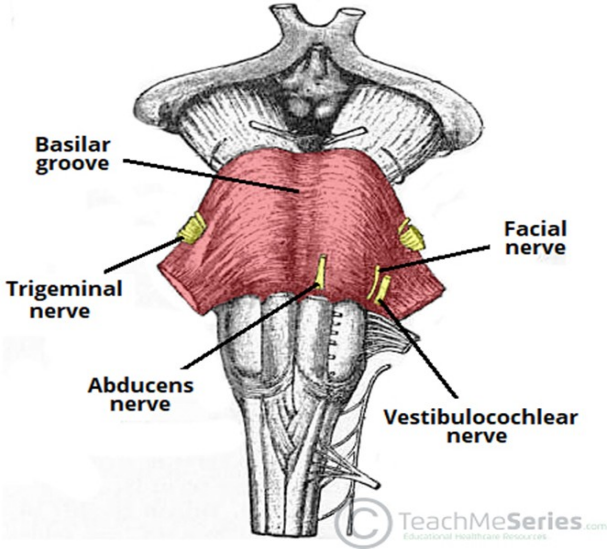
Course:

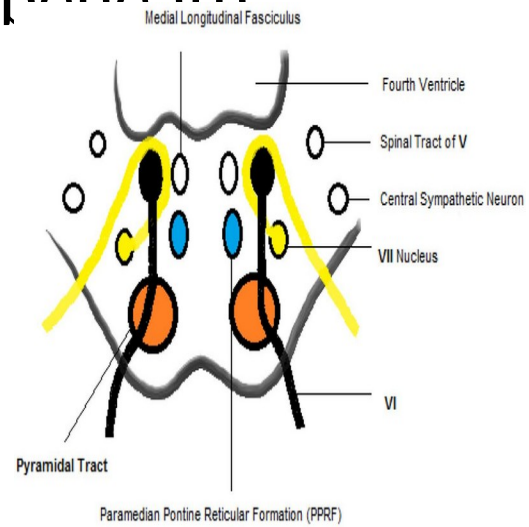
- It is the only nerve which arise from dorsal aspect of brainstem.
- It winds round superior cerebellar peduncle and cerebral peduncles to reach the ventral surface between temporal lobe and upper border of pons.
- Then pierces the roof of cavernous sinus and runs in the lateral wall between oculomotor and ophthalmic nerves.
- The nerve enters orbit through superior orbital fissure and ends by supplying superior oblique muscle.
- **Applied anatomy:** When trochlear nerve is damaged, *diplopia* occurs on looking downwards.

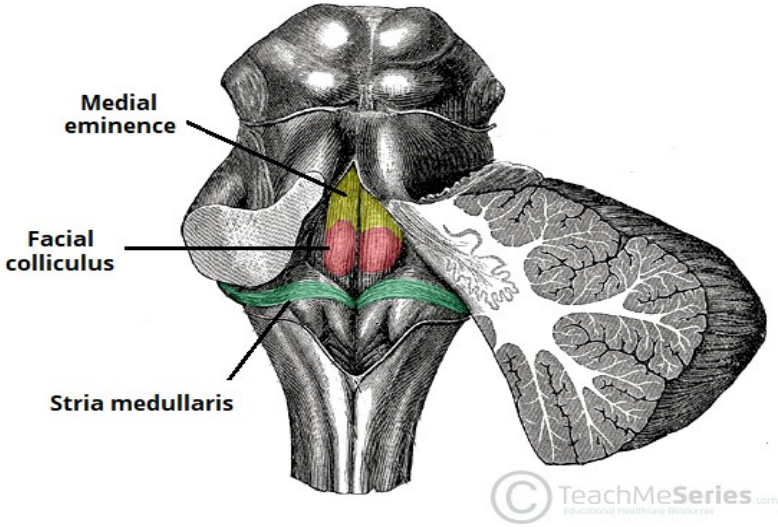


Abducent nerve (6th):

- It is motor and supplies lateral rectus muscle.
- **Nucleus:** abducent nucleus situated in the upper part of floor of fourth ventricle beneath facial colliculus.







Course:

- The nerve is attached to lower border of pons opposite upper end of pyramid.
- It enters the cavernous piercing its roof and passes through its center lateral to internal carotid artery.
- It enters orbit passing through superior orbital fissure and ends by supplying only lateral rectus muscle.
- **Applied anatomy:** Damage to abducent nerve results in:

a) Internal/medial squint, &

b) Diplopia

